
VANGUARD-32

Autonomous Smart Perimeter Defense & Active Countermeasure Platform

ATmega32-Based Embedded Systems Design
Project Proposal

1. Motivation

Problem Description & Importance

Problem Description

- Conventional installations rely on a single sensor trigger (PIR, optic tripwire).
- False alarms from wind, foliage, and small animals; no automated local verification.
- Passive design: sensors cannot re-orient toward an intrusion sector or discriminate personnel from intruders.

Why It Matters

- Critical infrastructure, remote outposts, and industrial zones need target acquisition and verification without human latency.
- Reducing false alarms while keeping detection reliability is a core embedded-defense challenge.
- A multi-stage verification pipeline is required before an alarm or countermeasure fires.

1. Motivation

Why This Project & Main Objectives

The project integrates multi-sensor fusion, closed-loop actuation, multi-bus digital communication, and low-latency interrupt handling on a resource-constrained 8-bit MCU (ATmega32) – a complete embedded system, not an isolated sensor demo.

1. **Multi-sector triangulation:** 180° field split into 3 sectors, hardware interrupts.
2. **Pan-tilt acquisition:** 2-axis turret aligns to the target vector within 500 ms via hardware PWM.
3. **Multi-modal verification:** thermal ΔT (MLX90614) and ToF distance R (VL53L0X) over I²C.
4. **Wireless IFF handshake:** encrypted RF badge authentication over SPI to prevent friendly engagement.
5. **Active countermeasure:** strobe pulse / solenoid actuation on confirmed hostile target.
6. **Telemetry & black-box logging:** live radar view over UART; on-board MicroSD incident log.

2. Equipment

Hardware Components

- **ATmega32-16PU** – system controller
- **RCWL-0516 / PIR array** – sector detection (INT0/1/2)
- **MLX90614** – IR thermal sensor (I²C)
- **VL53L0X** – laser ToF distance (I²C)
- **nRF24L01+** (×2) – 2.4 GHz IFF link (SPI)
- **MicroSD module** – black-box logging (SPI)
- **SG90 / MG996R servos (×2)** – pan-tilt turret (PWM)
- **FT232RL** – USB-UART telemetry bridge, 115200 baud
- **MOSFET drivers + solenoid/strobe** – countermeasure output
- **16×2 I²C LCD + LM2596 buck converters (×2)**

2. Equipment

Why These Components

Sensing & Detection

- **ATmega32:** on-chip TWI, SPI, USART, and two 8-bit timers cover every required bus and PWM channel without external bridge chips.
- **RCWL-0516/PIR:** outputs a digital edge directly on an interrupt pin – sub-ms detection with zero MCU polling.
- **MLX90614:** factory-calibrated digital I²C output (addr. 0x5A) – no analog front end or ADC channel needed.
- **VL53L0X:** laser ToF gives direct mm-accurate range (addr. 0x29) on the same I²C bus, avoiding ultrasonic beam-spread error.

Actuation, Comms & Power

- **Servos:** accept a standard 50 Hz PWM signal directly from Timer0/Timer2 OC pins – no motor driver IC required.
- **nRF24L01+ / MicroSD:** both are SPI slaves, sharing one hardware bus via independent chip-select lines instead of two controllers.
- **FT232RL:** the ATmega32 has no native USB peripheral; this bridges USART to a PC USB port.
- **MOSFET + solenoid/strobe:** MCU I/O pins cannot source actuator current; the MOSFET is a low-side switch driven by one GPIO.
- **Dual LM2596 rails:** isolates noisy servo current draw from the logic supply, preventing brown-out resets.

2. Equipment

Software & Development Tools

Toolchain

- **Microchip Studio / AVR-GCC** – C development, register-level drivers, compilation
- **USBasp / eXtreme Burner** – firmware flashing
- **Proteus VSM** – circuit simulation before fabrication

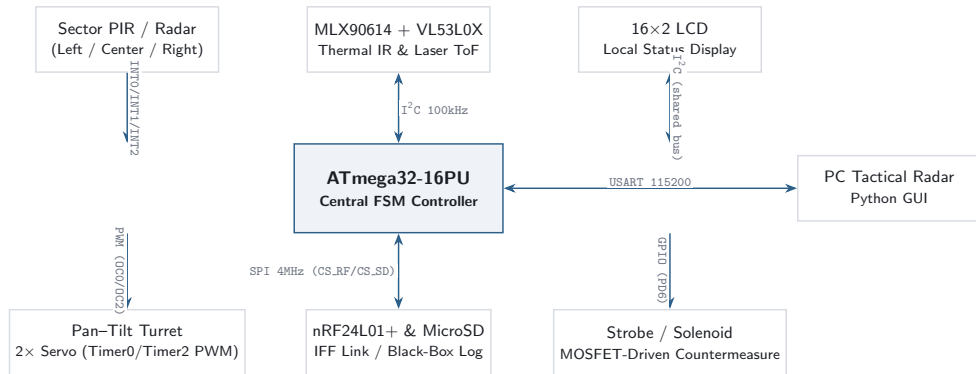
Host-Side Tools

- **Python 3 (PySerial)** – UART telemetry parsing
- **Pygame / Tkinter** – real-time PC tactical radar GUI

Sensors provide raw data; the ATmega32 fuses it and drives the response; host tools give visibility only – the MCU operates autonomously without the PC link.

3. System Design

Block Diagram



→ one-way signal ↔ shared/bidirectional bus – ATmega32 is the sole arbitration point for all three buses (interrupt lines, I²C, SPI) and the PWM/USART/GPIO peripherals.

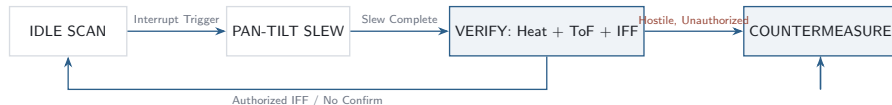
3. System Design

ATmega32 Interface Pinout Allocation Map

Pin(s)	Function	Connected To
PD0 (RXD), PD1 (TXD)	Hardware USART	FTDI FT232RL, 115200 baud
PD2 (INT0)	External interrupt 0	Left sector sensor trigger
PD3 (INT1)	External interrupt 1	Center sector sensor trigger
PB2 (INT2)	External interrupt 2	Right sector sensor trigger
PB3 (OC0)	Timer0 hardware PWM	Pan servo control signal
PD7 (OC2)	Timer2 hardware PWM	Tilt servo control signal
PC0 (SCL), PC1 (SDA)	Hardware TWI (I ² C)	MLX90614, VL53L0X, 16×2 LCD (shared)
PB5/PB6/PB7 (MOSI/MISO/SCK)	Hardware SPI	nRF24L01+ and MicroSD (shared)
PD4	GPIO output	MicroSD chip select (CS_SD)
PD5	GPIO output	nRF24L01+ chip select (CS_RF)
PD6	GPIO output	Countermeasure MOSFET gate
PA0 (ADC0)	Hardware ADC channel 0	Battery voltage monitor

4. Role of the ATmega32

Finite State Machine Logic



The

ATmega32 acts as the central real-time FSM: signal acquisition, bus arbitration, timing, and peripheral management all run on a single controller.

4. Role of the ATmega32

Peripheral Utilization Breakdown

- **GPIO:** MOSFET power stages, status indicators, SPI chip-select lines.
- **External interrupts (INT0/1/2):** sub-millisecond intrusion detection across 3 sectors, no polling overhead.
- **Timer0 & Timer2 (8-bit):** Phase-Correct/Fast PWM, 50 Hz on OC0 and OC2 for dual-axis servo positioning.
- **Timer1 (16-bit):** microsecond system clock for non-blocking task scheduling.
- **Hardware TWI (I²C):** 100 kHz bus to MLX90614 (0x5A), VL53L0X (0x29), and local LCD.
- **Hardware SPI:** 4 MHz master mode, multiplexed between nRF24L01+ and MicroSD via software CS.
- **Hardware USART:** 115200 baud, continuous telemetry frames (\$ANGLE, DIST, TEMP, STATUS#).
- **ADC:** reads supply voltage on ADC0 to detect low-power conditions.

5. Implementation Plan

Milestones

Project Update 1 (Wks 1–3)

- 1.1 Bare-metal clock, power, register setup
- 1.2 Timer0/Timer2 50 Hz PWM for dual-axis servos
- 1.3 Hardware I²C driver: MLX90614 + VL53L0X
- 1.4 External interrupts for sector detection

Deliverable: turret slews to activated sector, outputs distance/temperature over UART.

Final Demonstration (Wks 4–6)

- 2.1 SPI multiplexing (nRF24L01+ + MicroSD, independent CS)
- 2.2 Wireless IFF challenge-response handshake
- 2.3 Python PC tactical radar GUI over UART
- 2.4 Countermeasure MOSFET drive integration
- 2.5 Enclosure, power decoupling, stress test, tuning

Deliverable: full platform – detection, tracking, IFF, radar.

5. Implementation Plan

Group Member Responsibilities

Member 1: Firmware & Control

- Register-level drivers (Timers, PWM, I²C, ext. interrupts)
- Core FSM architecture
- Timing loop optimization

Member 2: Hardware, Power & Actuation

- Pan-tilt turret assembly
- Power distribution / decoupling
- Sensor calibration; MOSFET driver circuitry

Member 3: Wireless, Storage & GUI

- SPI bus mgmt. (nRF24L01+ + MicroSD)
- IFF protocol logic
- Python tactical radar GUI

6. Use of Other Controllers

Optional Auxiliary Node

Main System

The ATmega32 is the sole main microcontroller: all core logic, sensor readings, motor drives, storage logging, and telemetry streaming run on it.

Auxiliary Controller – Wearable IFF Badge

- **Device:** Arduino Nano / ATtiny85, battery-powered, worn by personnel.
- **Why required:** the badge is a mobile node distinct from the fixed turret station.
- **Tasks:** listens for a 2.4 GHz challenge packet from the ATmega32 via its own nRF24L01+ module and responds with a pre-shared 32-bit authorization hash.
- **Interface:** wireless RF only – performs no main system tasks.

7. Testing & Success Criteria

Measurable Criteria & Test Methodology

Metric	Target	Test Method	Priority
Turret slew speed	< 500 ms	Trigger-to-motion timing	Essential
Distance range & precision	± 5 mm at ≤ 1.5 m	VL53L0X bench test	Essential
Thermal verification	$\Delta T \geq 3^{\circ}\text{C}$ above ambient	Controlled heat-source test	Essential
IFF handshake execution	< 30 ms	RF range/response timing	Essential
PWM & turret accuracy	Stable 50 Hz on OC0/OC2	Oscilloscope waveform check	Essential
I ² C bus stability	Clean SCL @ 100 kHz	Multi-device signal integrity	Essential
Telemetry streaming rate	≥ 10 Hz, no frame loss	PC GUI continuous update	Optional
MicroSD black-box log	Event timestamp/angle/heat	File readback verification	Optional

Essential: multi-sector interrupt detection, dual-axis PWM tracking, I²C thermal/ToF check, basic countermeasure trigger. Optional: MicroSD logger, PC radar GUI, audio warning tone, ADC battery monitor.

8. Challenges & Future Scope

Anticipated Issues & Scaling Path

Technical & Hardware Challenges

- **SPI bus contention:** nRF24L01+ (low latency) vs. MicroSD (bulk writes) → dedicated CS lines, explicit bus-release routines.
- **Inductive noise:** servo current spikes risk brown-out → isolated 5V/6V buck rails, 1000 μ F decoupling.
- **Non-blocking I²C:** slow sensor response could stall main loop → state-driven polling, no blocking delays.

Future Scope

- **Computer vision:** ESP32-CAM co-processor for facial recognition / visual target classification.
- **360° slewing:** continuous stepper motors and slip rings for full perimeter coverage.
- **Long-range telemetry:** LoRa mesh for multi-kilometer, multi-unit perimeter networking.